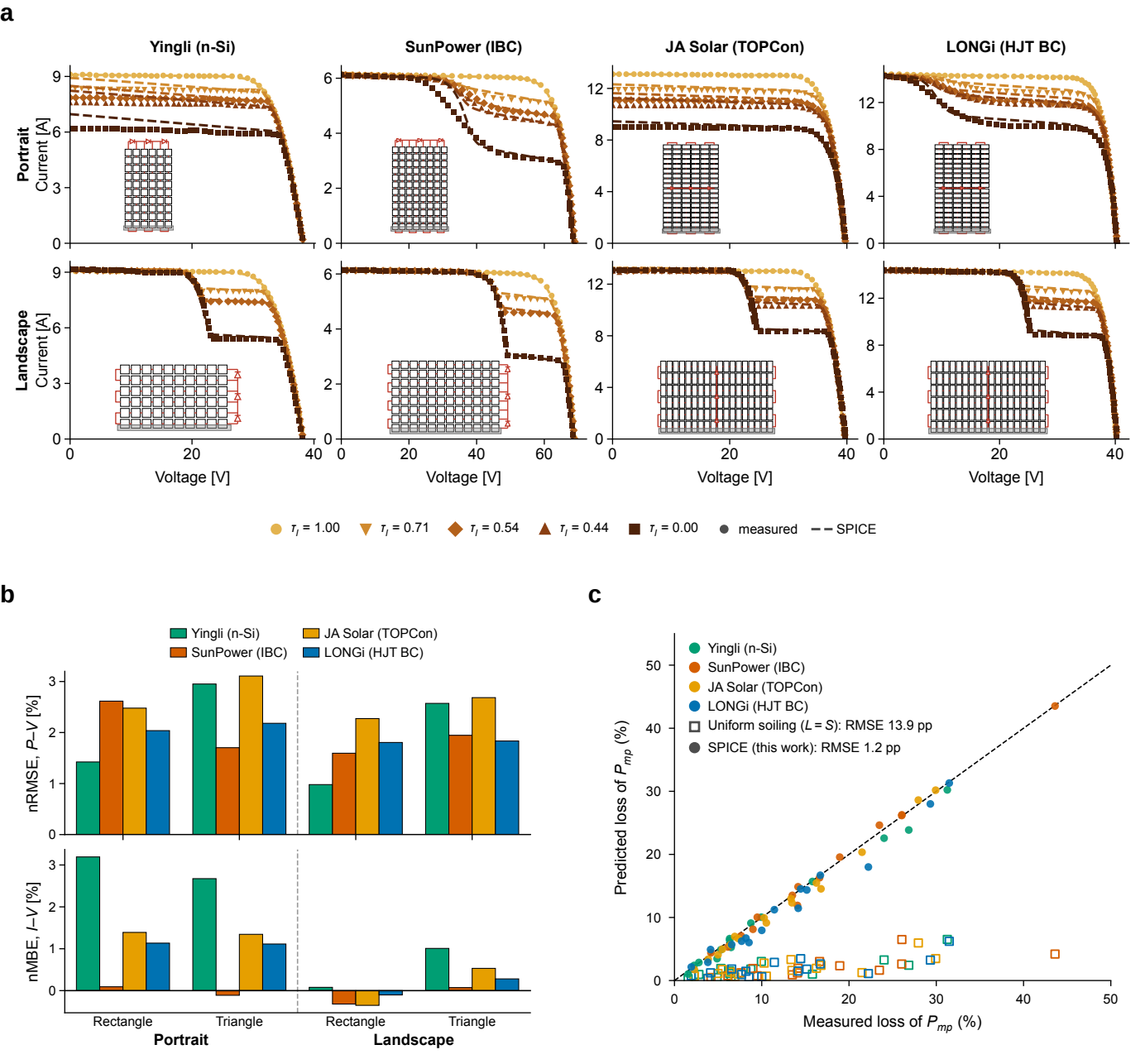

a
Yingli (n-Si)
SunPower (IBC)
JA Solar (TOPCon)
LONGi (HJT BC)
Portrait
Landscape
Current [A]
Voltage [V]
τ_l = 1.00
τ_l = 0.71
τ_l = 0.54
τ_l = 0.44
τ_l = 0.00
measured
SPICE
b
nRMSE, P–V [%]
nMBE, I–V [%]
Rectangle
Triangle
Portrait
Landscape
c
Predicted loss of P_{mp} (%)
Measured loss of P_{mp} (%)
Uniform soiling (L = S): RMSE 13.9 pp
SPICE (this work): RMSE 1.2 pp